Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2008-II-Q05] | Part A (Q1-Q5)

Trapezoid $ABCD$ has $BC \parallel AD$, $BC = 1000$, $AD = 2008$, and base angles $A = 37^\circ$, $D = 53^\circ$. If M, N are the base midpoints, find MN.

梯形 $ABCD$ 中 $BC \parallel AD, BC = 1000, AD = 2008$，底角 $A = 37^\circ, D = 53^\circ$。$M, N$ 为两底中点，求 MN。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2009-I-Q05] | Part A (Q1-Q5)

Triangle ABC has $AC = 450, BC = 300$. Let K be the midpoint of AC, CL the internal angle bisector meeting AB, and $P = BK \cap CL$. Reflect P across K to point M. Given $AM = 180$, find LP.

三角形 ABC 中 $AC = 450, BC = 300$，K 为 AC 中点，内角平分线 CL 交 AB 于 L，$P = BK \cap CL$。将 P 关于 K 对称得 M，若 $AM = 180$，求 LP。

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Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2016-II-Q05] | Part A (Q1-Q5)

Right triangle ABC_0 has right angle at C_0 and pairwise coprime positive integral sides, with perimeter p. Let C_1 be the perpendicular foot from C_0 to AB. Recursively, C_n is the foot from C_{n-1} to $C_{n-2}B$ for $n \geq 2$. If $\sum_{n\geq 1} C_{n-1}C_n = 6p$, find p.

直角三角形 ABC_0 在 C_0 处为直角，三边为两两互质的正整数，周长为 p。C_1 为 C_0 到 AB 的垂足；$n \geq 2$ 时，C_n 为 C_{n-1} 到 $C_{n-2}B$ 的垂足。若 $\sum_{n\geq 1} C_{n-1}C_n = 6p$，求 p。

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Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2024-II-Q05] | Part A (Q1-Q5)

A convex hexagon has six equal sides and parallel opposite sides. The lines containing alternating sides AB, CD, EF form a triangle with sides $200, 240, 300$. Determine the hexagon side length.

凸六边形六边等长，三对对边分别平行。交替三边 AB, CD, EF 的所在直线围成边长为 $200, 240, 300$ 的三角形。求六边形边长。

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Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2001-I-Q07] | Part B (Q6-Q10)

In ABC with $AB = 21, AC = 22, BC = 20$, the line through the incenter parallel to BC cuts AB, AC at D, E. If $DE = m/n$ reduced, find $m + n$.

三角形 ABC 中 $AB = 21, AC = 22, BC = 20$。过内心且平行 BC 的直线交 AB, AC 于 D, E。若 $DE = m/n$ 已约分，求 $m + n$。

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Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2004-I-Q09] | Part B (Q6-Q10)

A side-$3, 4, 5$ triangle is cut into a triangle U_1 and trapezoid V_1. A 6×7 rectangle is likewise cut into a triangle U_2 and trapezoid V_2, with $U_1 \sim U_2$ and $V_1 \sim V_2$. If the minimum possible $[U_1] = m/n$ reduced, find $m + n$.

边长 $3, 4, 5$ 的三角形被一线段切为三角形 U_1 与梯形 V_1，6×7 矩形也被切为 U_2, V_2，并要求两三角形相似、两梯形相似。若 U_1 最小面积约分为 m/n，求 $m + n$。

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Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2015-II-Q07] | Part B (Q6-Q10)

Triangle ABC has $AB = 12, AC = 17, BC = 25$. A rectangle $PQRS$ has P on AB, Q on AC, and R, S on BC, with $PQ = w$. Its area is $\alpha w - \beta w^2$. If $\beta = m/n$ in lowest terms, find $m + n$.

三角形 ABC 中 $AB = 12, AC = 17, BC = 25$。内接矩形 $PQRS$ 的 P, Q 分别在 AB, AC 上，R, S 在 BC 上，$PQ = w$。面积为 $\alpha w - \beta w^2$。若 $\beta = m/n$ 为最简分数，求 $m + n$。

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Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2018-II-Q07] | Part B (Q6-Q10)

Triangle ABC has $AB = 9, AC = 12, BC = 5\sqrt{3}$. Parallel cuts $P_kQ_k \parallel BC$ divide it into 2450 equal-area bands, with $P_0 = Q_0 = A, P_{2450} = B, Q_{2450} = C$. Count rational lengths among P_kQ_k, $1 \le k \le 2450$.

三角形 ABC 中 $AB = 9, AC = 12, BC = 5\sqrt{3}$，用平行于 BC 的 P_kQ_k 将其分成二千四百五十块等面积区域，$P_0 = Q_0 = A, P_{2450} = B, Q_{2450} = C$。求 $1 \le k \le 2450$ 的截线中长度为有理数的个数。

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Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2019-I-Q06] | Part B (Q6-Q10)

In convex quadrilateral $KLMN$, $MN \perp KM$, $KL \perp LN$, $MN = 65, KL = 28$. The line from L perpendicular to KN meets diagonal KM at O, with $KO = 8$. Find MO.

凸四边形 $KLMN$ 中 $MN \perp KM, KL \perp LN, MN = 65, KL = 28$。从 L 作 KN 垂线交对角线 KM 于 O，$KO = 8$，求 MO。

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Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2019-II-Q07] | Part B (Q6-Q10)

Triangle ABC has $AB = 120, BC = 220, CA = 180$. Three lines parallel respectively to BC, CA, AB cut interior segments of lengths $55, 45, 15$. Find the perimeter of the triangle formed by these lines.

三角形 ABC 边长为 $AB = 120, BC = 220, CA = 180$。三线分别平行于 BC, CA, AB，在原三角形内截段长为五十五、四十五、十五。求三线围成三角形的周长。

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Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2001-II-Q13] | Part C (Q11-Q15)

Quadrilateral $ABCD$ satisfies $\angle BAD = \angle ADC$ and $\angle ABD = \angle BCD$, with $AB = 8, BD = 10, BC = 6$. If $CD = m/n$ reduced, find $m + n$.

四边形 $ABCD$ 满足 $\angle BAD = \angle ADC, \angle ABD = \angle BCD$，且 $AB = 8, BD = 10, BC = 6$。若 $CD = m/n$ 已约分，求 $m + n$。

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Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2002-II-Q13] | Part C (Q11-Q15)

In ABC, $AB = 8$, $CD = 2, DB = 5$ with $D \in BC$, and $CE = 1, EA = 3$ with $E \in AC$. Put $P = AD \cap BE$. Lines through P parallel to CA, CB meet AB at Q, R. Write $[PQR]/[ABC] = m/n$ reduced and find $m + n$.

三角形 ABC 中 $AB = 8$，$D \in BC$ 且 $CD = 2, DB = 5$，$E \in AC$ 且 $CE = 1, EA = 3$。令 $P = AD \cap BE$，过 P 分别作 CA, CB 平行线交 AB 于 Q, R。若面积比约分为 m/n，求 $m + n$。

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Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2004-II-Q13] | Part C (Q11-Q15)

Convex pentagon $ABCDE$ satisfies $AB \parallel CE, BC \parallel AD, AC \parallel DE$, with $\angle ABC = 120^\circ$, $AB = 3, BC = 5, DE = 15$. If $[ABC]/[EBD] = m/n$ reduced, find $m + n$.

凸五边形满足 $AB \parallel CE, BC \parallel AD, AC \parallel DE$，且 $\angle ABC = 120^\circ, AB = 3, BC = 5, DE = 15$。若面积比 $[ABC]/[EBD] = m/n$ 已约分，求其和。

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Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2006-II-Q12] | Part C (Q11-Q15)

Equilateral ABC has circumradius two. Extend AB, AC past B, C to D, E with $AD = 13, AE = 11$. Complete parallelogram $ADFE$. Line AF meets the circle again at G. If $[CBG] = p\sqrt{q}/r$ is reduced with squarefree q, find $p + q + r$.

等边三角形 ABC 外接圆半径为二，延长 AB, AC 至 D, E，使 $AD = 13, AE = 11$，补成平行四边形 $ADFE$。AF 与圆再交于 G。若 CBG 面积为约简的 $p\sqrt{q}/r$，求 $p + q + r$。

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Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2007-I-Q15] | Part C (Q11-Q15)

In equilateral ABC, $F \in AB, D \in BC$ satisfy $AF = 5, CD = 2$. A point $E \in CA$ has $\angle DEF = 60^\circ$ and $[DEF] = 14\sqrt{3}$. The two possible side lengths are $p \pm q\sqrt{r}$ with rational p, q and squarefree integer r. Find r.

等边三角形 ABC 内，$F \in AB, D \in BC$ 满足 $AF = 5, CD = 2$。$E \in CA$ 使 $\angle DEF = 60^\circ$，面积 $[DEF] = 14\sqrt{3}$。原三角形边长两个可能值为有理数 p, q 构成的 $p \pm q\sqrt{r}$，r 无平方素因子，求 r。

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Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2013-I-Q13] | Part C (Q11-Q15)

Begin with triangle AB_0C_0 having sides $12, 17, 25$ in that order. For each $n \geq 1$, take B_n on AB_{n-1} and C_n on AC_{n-1} so that the ordered triangles AB_nC_n, $B_{n-1}C_nC_{n-1}$, $AB_{n-1}C_{n-1}$ are similar. The union of triangles $B_{n-1}C_nB_n$ has reduced area p/q. Find q.

从三角形 AB_0C_0 开始，$AB_0 = 12, B_0C_0 = 17, C_0A = 25$。对每个 n，在 AB_{n-1}, AC_{n-1} 上分别取 B_n, C_n，使有序三角形 $AB_nC_n, B_{n-1}C_nC_{n-1}, AB_{n-1}C_{n-1}$ 相似。所有 $B_{n-1}C_nB_n$ 的并集面积为最简分数 p/q，求 q。

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